SignQuest App Proposal Written Report

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1 Introduction

Accessibility is needed for many groups of people who have a type of disability, including the deaf community and individuals who are hard of hearing. Their ways of overcoming this issue are through subtitles that are integrated in audio sources and communicating through a sign language standard, like American Sign Language (ASL). Close family and friends can communicate sign language with deaf people, by educating themselves with courses, lessons and videos. But for people who are introduced to any deaf individuals and do not have any experience with sign language, they might have trouble with communication, only speaking through written/typed notes or pointing to their target of discussion. Having an application that deals with this problem through educating and motivating can close the gap in communication with the deaf community. Recently, educational technologies have been used increasingly to support independent learning through interactive and accessible platforms. Language learning apps, like Duolingo, show how structured lessons with engagement-focused designs can improve retention and learner motivation. However, many language learning apps are designed for spoken languages and don't address the visual/motion-based parts of ASL.

Learning ASL requires users to develop mental understanding and physical skills, which cannot be achieved effectively from just reading about it. Beginners require repeated exposure, guided practice and feedback to recreate signs and understand the conversational context. Without the tools that encourage learners to engage in continuous practice, they may struggle to apply the skills in a real-world scenario.

This project will introduce an application that our group is making, called “SignQuest”, which is an educational web application aimed at supporting ASL beginners. Through interactive lessons, practice activities and gamified experiences, it will combine the educational structure with engagement design principles, so there is an accessible learning environment that promotes practice and improves communication between hearing individuals and the deaf community.

1.1 Problem

The learning curve for American Sign Language is one of the most difficult for beginners, as it involves learning a language in a new manner with limited tools and resources to do so. Unlike spoken languages, ASL is dependent on coordinated handshape, movement and facial expressions to create meaning. Beginners have trouble recognizing and accurately reproducing signs due to the lack of guided feedback and practice opportunities. Existing ASL learning resources are commonly limited to videos, classroom instructions or dictionary-style applications that focus only on memorization rather than skill development. The existing tools don't give an interactive practice environment that supports repeated learning attempts. Learners might experience little motivation and have difficulty when applying signs in real-world communication scenarios. The missing tools of accessible, structured and practice-oriented learning platforms create a barrier for individuals attempting to learn ASL independently.

1.2 Motivation

Communication between hearing individuals and members of the deaf community remains difficult in everyday social environments. While resources for learning ASL exist, many are structured for long-term formal studies or don't have engagement for beginner learners. This makes individuals who want to learn basic ASL for practical communication struggle and not progress beyond the introductory level. The motivation for creating this application is to have an accessible and engaging learning platform that encourages consistent practice. By combining educational principles with engagement design strategies, learners can develop practical communication skills that improve real-world interactions with the deaf community.

1.3 Our Solution

The application presented in this proposal, SignQuest, is an educational web platform that is designed to support beginner ASL learners through structured lessons, interactive exercises and gamified learning experiences. The application is focused on teaching commonly used signs, simulated conversations and recognition skills that can be applied to real-life communication scenarios. SignQuest incorporates gamification elements like progress tracking, achievements and leaderboards to encourage practicing and engagement. The app also requires active participation through visual demonstrations, step-by-step lessons and competing with other people. This approach can make ASL learning more approachable and motivating for new learners.

1.4 Target Audience

The primary target audience for SignQuest includes hearing individuals with little to no prior experience in American Sign Language who want to develop foundational communication skills. This includes students, educators, service workers, family members of deaf individuals and the general public who are seeking inclusive communication abilities. Secondary users include beginner ASL learners who are looking for practice outside of formal courses. The application is designed for users who want flexible and self-paced learning that fits into their daily routines while being accessible to users with varying levels of technological experience.

2. Background (with ACM format and references)

2.1 Digital Language Learning

The use of technology to learn language has been gaining popularity due to its accessibility, flexibility, and interactive features, with notable platforms like Duolingo and Babbel becoming household tools for language learning. The success of these applications is strongly associated with not online accessibility but with the ability to cover the three pillars of UDL: engagement, representation, and action. A recent study found that the greatest improvement was observed in vocabulary, reading, and listening skills in language-learning applications [1]. These results stem from the ability to receive instant feedback and guided, repeated practice, as seen in Duolingo, and are commonly not available without technologies beyond the classroom.

Time spent in learning a language is not the only factor that leads to success, but also the content methodology, task structure and level of interactivity, also discussed in gamification and correlated with the UDL pillars [1]. Ensuring that a variety of content is available to users across different learning objects is important to users' success. A literature review found that technology-based language learning not only improves language skills but also metacognitive skills, positive attitudes towards learning, language learning retention, and more [2]. These benefits associated with digital language learning are interdependent on the repetition and mechanisms involved in learning a language, leading to higher language proficiency for users.

2.2 ASL Challenges

Learning ASL poses unique challenges compared to learning other languages due to its visual and physical aspects. There are three functional movements involved in ASL, as well as parameters that aid in phonological awareness (the ability to manipulate the sound structure of languages) [3]. See Table A.

These different actions and parameters involve expressing not only unique words but also sentence structures, such as asking questions or making statements. Grammar also needs to take into account, and there are three important concepts to understand:

- Morphology: how the words are grammatically formed [3].
- Semantics: the meaning of the words and context [3].
- Syntax: the rules of forming sentence structure [3].

These grammar rules are expressed through the different parameters of sign language. However, it is important to note that,t similar to other spoken languages, ASL has regional and sociological accents and dialects [4]. This makes creating an ASL application extremely difficult, as sociological and regional biases might not be accounted for (difficult for users trying to communicate with people who use these dialects) due to the significant resources required to implement all accents and dialects fully.

Furthermore, parameters and functional movements can be broken into different linguistic categories of producing signs. These categories usually involve different movements for different grammatical implementations, shortcuts or spelling. The following are the main linguistic categories:

- Fingerspelling: spelling words with the alphabetical hand signs. [3, 4]
- Lexical Signs: a fixed sign for one word [3, 4]
- Classifiers: handshapes representing nouns [4]
- Non-manual markers: provide context to the sign through action [4]

The implementation of SignQuest, especially its camera translation capability, will need to be scaled back from a full implementation of ASL elements to a beginner version. The scale-down is due to the significant resources required to fully implement an ASL learning application without its full language dynamics and constraints. A focus on fingerspelling, lexical signs with hand movements and syntax will be implemented. At the same time, morphology and semantics will be briefly covered, without implementing full-body and face detection. These different functional movements and concepts can be covered conceptually, but are difficult to implement in an ASL detection model.

2.3 Gamification

Gamification is the integration of game design principles, such as rewards, leaderboards, badges, challenges, and more, into applications to increase engagement, retention, and academic performance [5]. However, the use of gamification in language learning technologies and its effectiveness are unique compared to its use in other educational subjects. To successfully implement gamification, understanding the learning objectives of most language applications is crucial. These learning objectives prioritize vocabulary, sentence structure and grammar, with less popular objectives involving pronunciation and reading comprehension [6]. Language applications can leverage gamification features to achieve these learning objectives. Empirical studies have shown that gamification directly correlates with engagement, with increased behavioural, emotional, and cognitive engagement, all of which are crucial for learning skills that require extensive practice, such as language learning [6]. Thus, the outcomes of gamification features help achieve learning objectives.

An important consideration is the possibility of demotivation from gamification [6]. Users may be discouraged or overwhelmed by competitive features like leaderboards or badges if they are disappointed with their results. On the other hand, if the results of leaderboards, badges or other gamification features are related to a user's struggle in a particular field, these features can be used as a learning analytics to indicate to the user and systems which fields need improvement [5].

SignQuest will use gamification features to aid in progress tracking and encourage confidence and repetition, all of which are crucial when developing language skills.

2.4 Technologies

Given the importance of interactivity and diverse learning objects, SignQuest will need to leverage real-time computer vision techniques to detect ASL signs. In terms of the ASL detection model, the tools in Table B will be used based on the paper "Real-Time American Sign Language Interpretation Using Deep Learning and Keypoint Tracking" [7]

The process begins with OpenCV accessing the camera. Then the MediaPipe BlazePalm detector finds the hands in each frame. Next, YOLO (You Only Look Once) is used to detect and classify 21 keypoints, a model commonly used for real-time object detection [7]. Given the use of live translation in many learning objects, a fast and reliable model is crucial. This model will focus solely on finding the hands and their orientation.

3. App Design

3.1 Learning Objectives

The educational foundation of SignQuest is built on SMART Learning Objectives to ensure a structured, effective learning experience. The app defines specific goals where users learn the twenty-six letters of the ASL alphabet and over 100 core signs. Success is measurable using the integrated camera model, which provides percentage-based accuracy for handshape and motion, ensuring the criteria are met when assessing success. These objectives are achievable because the challenges are divided into easy tasks, such as individual letters and basic signs; medium tasks, including more advanced signs; and hard tasks, such as more advanced sign combos and more complete conversations in ASL, matching the users' increasing skill level as they progress through the app. The curriculum is highly relevant, as it connects signs to real-life daily interactions through scenarios such as "Ordering Coffee" and "Asking for Directions." Finally, the app offers time-bound lessons in the form of microlearning modules, ranging from 10 to 22 minutes, to help users manage their time and maintain motivation.

3.2 Learning Objects

SignQuest follows the LEGO building-block analogy by treating each educational unit as a small, reusable Learning Object that can be combined to produce customized learning experiences. Each Learning Object is a self-contained activity designed to meet one specific learning objective, typically lasting 5 to 12 minutes. An example of a Learning Object in the app is the Alphabet Video in a lesson, which provides a focused demonstration of handshapes for letters A through Z in ASL. The app also uses Simulation Scenario Learning Objects, such as the "Ordering Coffee" module, to act as an assessment tool for greetings and food-related signs.

3.3 Core Features

The core features of SignQuest are designed to balance user experience with learner efficacy. The Camera Practice provides immediate feedback and visual cues, such as bells or dings, which act as positive reinforcement to strengthen engagement. The Sign Combos feature is a game in which learners combine base signs to discover complex phrases, encouraging intrinsic motivation through curiosity rather than just earning basic points. Competitive elements are introduced through the Sign Battle mode, which utilizes leaderboards to create social motivation. To provide visible progress, the app uses a system of badges and achievements that encourage continued participation and give users a sense of growth and accomplishment.

3.4 Storyboard

When first storyboarding for SignQuest, we focused on two perspectives. First, a new user signing up for their first lesson and an experienced user playing a built-in game. Our original storyboard was designed for a mobile application, while the second iteration of storyboards was for a web application. From these two perspectives, we mapped the navigation from initial onboarding, where users learn the basics and navigate SignQuest, to an experienced user taking advantage of everything SignQuest has to offer.

3.5 UI/UX

The UI/UX design of SignQuest follows senior design principles, prioritizing functional learning over flashy effects. The colour palette adheres to the 60/30/10 rule, using 60% neutral tones, 30% complementary tints, and 10% brand colour to avoid a distracting or overwhelming learning experience. The app's structure is built on a consistent 8-point grid system to ensure professional, deliberate spacing. Copywriting is kept clear by using clear labels to reduce clutter and guide the user effectively. To support diversity, the app uses Universal Design for Learning strategies, providing high-contrast colours, readable fonts, and clear layouts. The app also provides Learning Objectives in visual and auditory formats to ensure accessibility for all users.

3.6 Technical Design

The technical design focuses on desirability, feasibility, and user success. While most UX designs typically aim to reduce effort, SignQuest's technical design requires 'Productive Struggle' during camera practice to improve learning outcomes. The plan for incorporating our design involves a robust framework. OpenCV provides the necessary access to the camera, while MediaPipe detects and tracks hand movements. Within this framework, the BlazePalm Detector identifies the palm and establishes a bounding box, enabling the Hand Landmark Model and a pretrained Convolutional Neural Network to extract features from 21 keypoints. Finally, a YOLO Model is implemented to detect and classify specific ASL letters and words using these landmarks. This design is intended to provide a safe environment to fail, where users receive immediate responses to their actions.

4. Conclusion

SignQuest brings together important teaching principles and advanced technology to help beginners learn ASL more easily. We built the app using the three main components of UDL: engagement, representation, and action to turn a boring dictionary into a fun, interactive environment. Our technical architecture uses a combination of OpenCV, MediaPipe, and YOLO models to give users instant feedback. This creates a productive struggle that helps people actually remember and understand how to sign correctly. By using SMART learning objectives and small learning objects, we make sure the tricky parts of ASL, like handshape, movement, and location, are easy to learn and track.

We also added fun gamification features, such as Sign Battle and Sign Combos, to make practicing ASL feel like an exciting game rather than a boring chore. We balanced these games and the rest of the app with senior UI/UX design rules to keep the screen simple and easy to use. As more people learn sign language online, SignQuest offers a unique way to master the visual aspects of sign language. By providing users with a safe environment to fail and a reason to keep practicing, our app helps people develop the skills needed to communicate with the deaf community in ASL.

Table A:

Functional Movements	Parameter	How it works
Hand Movements	Handshape	Shape of the Hand
	Movement	The action or path the hand takes.
	Location	Where the sign is made with respect to your body and air.
	Palm Orientation	The orientation of the palm when the sign is made
Body & Face	Non-Manual Markers	Facial grammar, as well as head and body shifts, made with the sign

Table B:

Tool	Purpose	How it works
OpenCV	Camera	Access to the camera
MediaPipe	Detect and tracking	Detects and tracks Hands
BlazePalm Detector	Detect Hand	Helps detect palm and bounding box
Hand Landmark Model	Feature extraction	With the YOLO model, this model is a pretrained CNN with 21 detected keypoints.
YOLO Model	Detect ASL	Classified ASL letters and words using landmarks

App Design (Storyboard + link to Figma)

Figma Links

(Version 1): Original Idea, less fleshed out
https://www.figma.com/site/X6Gx8tPsZE6vUS3HXijAgc/SignQuest---Beta-Design?node-id=0-1&t=M8Qrzj25eBlk3aiJ-1

(Version 2): Built upon version 1 with more detail, almost fully fleshed out idea
https://www.figma.com/design/zZE9LBRwPEHFRd8UbrPtRC/SignQuest---Design?node-id=0-1&t=rN53Q9IsXhtJAtlM-1

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